

Gaussian Trajectory Geometry: Reference Anisotropy, Occupied Frames, and Conditional Spectra

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We collect the Gaussian formalism behind three complementary numerical probes of an adaptive class-A fermion circuit. Pure trajectories are evolved as orthonormal occupied frames, while maximally mixed trajectories are retained as correlation matrices. For a fixed nonzero-probability measurement record, differentiation of the normalized correlation-matrix Möbius map gives an exact congruence cocycle. We prove that this cocycle is the Jacobian of the occupied-frame evolution after quotienting the frame's unitary gauge, and that its physical pure-state singular values are pairwise products of occupied and empty one-leg singular values. This identifies the invariant spectrum targeted by the pure-state G5 estimator and distinguishes it from the trajectory-operator spectrum reconstructed by the maximally mixed G4 calculation. We also give the charge-conserving Gaussian reference construction used to calibrate spacetime anisotropy and work through the completed coarse-audit matching estimator without promoting it to a final result.

I. GAUSSIAN COORDINATES AND NATIVE OCCUPIED FRAMES

For complex fermions satisfying $\{\hat{\psi}_i^\dagger, \hat{\psi}_j\} = \delta_{ij}$, the canonical correlation matrix is

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{\psi}_i^\dagger \hat{\psi}_j), \quad 0 \leq G \leq \mathbf{1}. \quad (1)$$

The implementation uses the transposed one-body convention $C = G^\top$ and the centered covariance

$$\Gamma = 2C - \mathbf{1}. \quad (2)$$

Transpose does not alter the spectra or Frobenius singular values discussed below. A pure charge-conserving Gaussian state obeys $C^2 = C$ and is represented by an orthonormal occupied frame $F \in \mathbb{C}^{N \times r}$,

$$\begin{aligned} F^\dagger F &= \mathbf{1}_r, & C &= F F^\dagger, \\ F &\sim F U, & U &\in U(r). \end{aligned} \quad (3)$$

Thus F is a Stiefel representative of a point on $\text{Gr}(r, N)$; the right-unitary freedom is numerical gauge, not a physical degree of freedom.

Let χ be a normalized measured orbital, $P_\chi = \chi \chi^\dagger$, and $Q_\chi = \mathbf{1} - P_\chi$. The occupied and empty Born probabilities are $n_\chi = \chi^\dagger C \chi$ and $1 - n_\chi$. The two primitive rank-changing frame operations have correlation matrices

$$C_{\text{gain}} = C + \frac{[(\mathbf{1} - C)\chi][(\mathbf{1} - C)\chi]^\dagger}{1 - n_\chi}, \quad (4)$$

$$C_{\text{loss}} = C - \frac{(C\chi)(C\chi)^\dagger}{n_\chi}. \quad (5)$$

Numerically, gain appends the normalized residual $(\mathbf{1} - C)\chi$ to F . Loss computes $a = F^\dagger \chi$, applies a complex Householder reflector that rotates a to the last frame coordinate, and deletes that column. An occupied projector is loss followed by deterministic gain; an empty projector is gain followed by deterministic loss. These compositions give

$$\begin{aligned} C_1 &= C - \frac{C P_\chi C}{n_\chi} + P_\chi, \\ C_0 &= C + \frac{(\mathbf{1} - C) P_\chi (\mathbf{1} - C)}{1 - n_\chi} - P_\chi. \end{aligned} \quad (6)$$

Thin QR is invoked only when the Gram residual warrants it. The CPU engine stores one variable-rank frame, whereas the GPU engine stores trajectory-dependent ranks in a padded batch and applies the same append/Householder algebra selectively. Frame-native trajectory evolution therefore avoids repeatedly materializing an $N \times N$ covariance. These identities are the frame-level realization of charge-conserving Gaussian measurements and feedforward [1, 2].

II. NORMALIZED MÖBIUS TANGENT AND THE FRAME JACOBIAN

Write the selected measurement outcome as $m \in \{0, 1\}$, $\eta_m = 2m - 1$, and the controller target as $s \in \{0, 1\}$. On the code-side centered covariance, one normalized measurement followed by reset is

$$\begin{aligned} \Phi_{m|s}(\Gamma) &= Q_\chi (\mathbf{1} + \eta_m \Gamma P_\chi)^{-1} \Gamma Q_\chi + \eta_s P_\chi, \\ p_m &= \frac{1 + \eta_m g}{2}, \quad g = \text{tr}(P_\chi \Gamma), \end{aligned} \quad (7)$$

For $p_m > 0$, the rank-one inverse is exact and

$$L_m(\Gamma) = Q_\chi (\mathbf{1} + \eta_m \Gamma P_\chi)^{-1} = Q_\chi - \frac{\eta_m Q_\chi \Gamma P_\chi}{1 + \eta_m g}. \quad (8)$$

Put $M = \mathbf{1} + \eta_m \Gamma P_\chi$. At fixed P_χ, m, s , the identity $\delta M^{-1} = -M^{-1}(\eta_m X P_\chi) M^{-1}$ gives

$$\begin{aligned} \delta \Phi &= Q_\chi M^{-1} X [\mathbf{1} - \eta_m P_\chi M^{-1} \Gamma] Q_\chi \\ &= L_m X L_m^\dagger, \end{aligned} \quad (9)$$

because $[\mathbf{1} - \eta_m P_\chi M^{-1} \Gamma] Q_\chi = (\mathbf{1} + \eta_m P_\chi \Gamma)^{-1} Q_\chi = L_m^\dagger$. Thus

$$\boxed{D_\Gamma \Phi_{m|s}[X] = L_m X L_m^\dagger.} \quad (10)$$

The reset term is constant, so its target s drops out. Exact projectors really are singular: $P_\chi L_m = 0$ and, for pure Γ , $\ker L_m = \text{span}\{(\mathbf{1} + \eta_m \Gamma)\chi\}$. This vector lies in the selected η_m eigenspace: that one-leg block loses one rank,

whereas the opposite block is injective. Such nulls are not pseudoinverted. A unitary event $\Gamma \mapsto UTU^\dagger$ has one-leg factor $L = U$. Writing L_t for either kind of event factor, a fixed ordered record gives

$$\begin{aligned} A_{T:0} &= L_{T-1} \cdots L_1 L_0, \\ X_T &= A_{T:0} X_0 A_{T:0}^\dagger. \end{aligned} \quad (11)$$

This is a branchwise derivative of the normalized conditional state, not a derivative through the outcome sampler and not a Born-score response.

We now make precise in what sense Eq. (11) is the frame Jacobian. Let $f_{m|s}$ denote the exact gain/loss frame update and $\pi_r(F) = 2FF^\dagger - \mathbf{1}$. If $r' = r + s - m$, the state-level square commutes:

$$\begin{array}{ccc} F & \xrightarrow{f_{m|s}} & F' \\ \pi_r \downarrow & & \downarrow \pi_{r'} \\ \Gamma & \xrightarrow{\Phi_{m|s}} & \Gamma' \end{array} \quad (12)$$

Equivalently, $\pi_{r'} \circ f_{m|s} = \Phi_{m|s} \circ \pi_r$; differentiation gives the commuting Jacobian relation

$$D\pi_{r',F'} \circ Df_{m|s,F} = D\Phi_{m|s,\Gamma} \circ D\pi_{r,F}. \quad (13)$$

Vertical variations $\delta F = F\Omega$, $\Omega^\dagger = -\Omega$, lie in $\ker D\pi_{r,F}$. A horizontal variation satisfies $F^\dagger \delta F = 0$ and

$$\begin{aligned} D\pi_{r,F}[\delta F] &= 2(\delta FF^\dagger + F\delta F^\dagger), \\ \|D\pi_{r,F}[\delta F]\|_F^2 &= 8\|\delta F\|_F^2. \end{aligned} \quad (14)$$

The same metric factor appears at input and output, including for rectangular, rank-changing branches. Hence the horizontal quotient derivative $\mathcal{H}_{F'} Df_{m|s,F}|_{\mathcal{H}_F}$ and the covariance derivative restricted to the pure-state tangent have identical singular values. A raw derivative of Householder or QR representatives also contains vertical gauge conventions and pivot choices and is not the invariant object in this statement.

For completeness, the one-leg factors also intertwine the initial and final pure covariances:

$$\boxed{\Gamma' L_m = L_m \Gamma.} \quad (15)$$

To verify this directly, put $\eta = \eta_m$, $d = 1 + \eta g$, and use the $P_\chi \oplus Q_\chi$ decomposition

$$\begin{aligned} \Gamma &= \begin{pmatrix} g & w^\dagger \\ w & B \end{pmatrix}, \\ Bw &= -gw, \quad w^\dagger w = 1 - g^2. \end{aligned} \quad (16)$$

Purity $\Gamma^2 = \mathbf{1}$ gives the two middle relations. Equations (7)–(8) become

$$\begin{aligned} L_m &= \begin{pmatrix} 0 & 0 \\ -\eta w/d & \mathbf{1} \end{pmatrix}, \\ \Gamma' &= \begin{pmatrix} \eta_s & 0 \\ 0 & B - \eta w w^\dagger/d \end{pmatrix}. \end{aligned} \quad (17)$$

Because $(B - \eta w w^\dagger/d)w = -\eta w$, multiplication proves Eq. (15). Thus L_m maps occupied and empty eigenspaces of Γ into the corresponding eigenspaces of Γ' , including exact nulls. The intertwining identity uses $\Gamma^2 = \mathbf{1}$ and need not hold for a mixed state.

III. G5: PURE-STATE PHYSICAL TANGENT SPECTRUM

At a pure state, a physical Hermitian tangent obeys $X^\dagger = X$ and $\{\Gamma, X\} = 0$. If U_o and U_e are complete occupied and empty frames, every such tangent is uniquely parameterized by a complex matrix K :

$$X = 2(U_o K U_e^\dagger + U_e K^\dagger U_o^\dagger). \quad (18)$$

Set $B_b = A_{T:0} U_b$ for $b \in \{o, e\}$. The intertwining identity keeps the two images in orthogonal final eigenspaces, and

$$X_T = 2(B_o K B_e^\dagger + \text{h.c.}). \quad (19)$$

With singular-value decompositions $B_b = W_b \Sigma_b R_b^\dagger$, vectorization gives

$$\text{vec}(B_o K B_e^\dagger) = (B_e^* \otimes B_o) \text{vec } K. \quad (20)$$

Therefore the physical tangent singular spectrum is the pair product

$$\begin{aligned} \sigma_{ij}^{\text{phys}} &= \sigma_i^o \sigma_j^e, \\ \lambda_{ij}^{\text{phys}} &= \frac{\log \sigma_i^o + \log \sigma_j^e}{T_{\text{obs}}}. \end{aligned} \quad (21)$$

Over the real Hermitian tangent space, the real and imaginary parts of each K_{ij} give two equal real singular directions; G5 stores the complex pair once.

The numerical G5 experiment implements the pair-product estimator (21) without forming an $N^2 \times N^2$ superoperator. Each trajectory begins from an independently seeded Haar-random, half-filled Slater frame and is evolved natively with batch size one. The tangent initializer forms and diagonalizes the restricted $FF^\dagger$ once at cycle zero to build complete occupied and empty bases; subsequent live updates are analytic and matrix-free. It QR-stabilizes the two blocks separately after every cycle while multiplying the corresponding triangular factors into scaled cumulative cores. Cycles $1, \dots, N_y$ align the frames; at N_y only the cores and log accumulators are reset, not the aligned bases. At the declared integer-rounded quarter-, half-, three-quarter-, and full-window checkpoints, direct SVDs give cumulative core singular values. All occupied–empty sums are formed before the 16 rates closest to zero are selected. Core values below threshold enter as $-\infty$; only finite pair rates are eligible. The observer records branch/Born minima and QR null masks; invalid branches deactivate, while core-SVD residual and physical-pair Gram tolerances are hard acceptance gates. Per-cycle QR diagonals are diagnostics, not substitutes for the finite-window core SVD. Exact null masks

are legitimate, not failures. After an exact null, however, a QR null-space completion supports the fresh-window physical interpretation only if the preserved blocks remain complete current eigenspace bases; a rigorous restart would rebuild them from the current frame. Production uses $N_x = 20$, $N_y = 20, 30, 40, 50, 60$, $T = 2N_y$, and 25 independent Born trajectories for each wall and matched-trivial case. No $\pm\epsilon$ trajectories, replayable record, Choi state, or Born-score tangent is generated.

IV. G4: MAX-MIX TRAJECTORY-OPERATOR SPECTRUM

G4 extracts a different object. Start from $\hat{\rho}_0 = \hat{\mathbf{1}}/2^{N_{\text{orb}}}$ and let $\hat{V}_{\mathbf{m}}$ be the many-body evolution operator (mEO) for one realized record. The normalized endpoint and its weight are

$$\hat{\rho}_{\mathbf{m}} = \frac{\hat{V}_{\mathbf{m}} \hat{V}_{\mathbf{m}}^\dagger}{Z_{\mathbf{m}}}, \quad Z_{\mathbf{m}} = 2^{N_{\text{orb}}} p_{\mathbf{m}}. \quad (22)$$

If ν_i are its natural occupations, Gaussian factorization gives every normalized many-body eigenvalue and absolute mEO singular value:

$$\lambda_{\mathbf{n}} = \prod_i \nu_i^{n_i} (1 - \nu_i)^{1-n_i}, \quad \sigma_{\mathbf{n}}(\hat{V}_{\mathbf{m}}) = \sqrt{Z_{\mathbf{m}} \lambda_{\mathbf{n}}}. \quad (23)$$

Define

$$e_i = \log \frac{1 - \nu_i}{\nu_i}, \quad n_i^{(0)} = \mathbf{1}_{\nu_i > 1/2}, \quad q_i = 1 - 2n_i^{(0)}, \quad \delta_i = \frac{|e_i|}{2}. \quad (24)$$

For a flipped subset $\mathcal{S}$, the tower consists of sums $\sum_{i \in \mathcal{S}} \delta_i$ sorted at fixed charge $q(\mathcal{S}) = \sum_{i \in \mathcal{S}} q_i$. Its leading level is

$$\log \sigma_0 = \frac{1}{2} \left[N_{\text{orb}} \log 2 + \log p_{\mathbf{m}} + \sum_i \log \max(\nu_i, 1 - \nu_i) \right]. \quad (25)$$

Exact caps $\nu_i = 0, 1$ have infinite flip cost and retain an occupied/empty orientation.

Numerically, G4 evolves centered max-mix covariances in GPU batches of at most five. The streaming observer records convergence and impurity summaries every cycle but performs a full natural-occupation eigen-decomposition only at declared early, logarithmic, and $N_y, 3N_y/2, 2N_y$ checkpoints. It stores per-cycle conditional log-probability increments, cumulative sums, and event counts; the cumulative $\log p_{\mathbf{m}}$ in Eq. (22) supplies $Z_{\mathbf{m}}$. The occupations define the full factorized Fock spectrum, but the implementation saves its leading scale and a low charge-resolved subset-sum tower, not an explicit

$2^{N_{\text{orb}}}$ list. Eigenpair and natural-orbital Gram residuals are acceptance gates. G4 uses the same sizes, duration, and 25-trajectory wall/control counts as G5, but saves no covariance history, tangent cocycle, Choi state, or replayable ordered-outcome record.

For a fixed max-mix record there is also an exact endpoint dictionary for the normalized half-Jacobian A :

$$AA^\dagger = 4C_T(1 - C_T) = \mathbf{1} - \Gamma_T^2, \quad j_i = 2\sqrt{\nu_i(1 - \nu_i)} = \text{sech } \delta_i. \quad (26)$$

It explains why max-mix occupations determine tangent magnitudes, but it does not equate G4 with G5: their initial states, Born measures, seeds, records, normalizations, and reported spectra differ.

V. REFERENCE ANCILLAS AND SPACETIME ANISOTROPY

The two-dimensional topological slab supplies one-dimensional domain-wall boundaries, whose low-energy theory is a 1 + 1-dimensional CFT: one edge coordinate y plus time. Here L_y is the circumference of one boundary. The transverse size N_x controls slab width and edge-edge coupling, not the CFT circumference; the two domain walls carry separate, oppositely chiral edge copies. For dynamical exponent $z = 1$, the expected scaling is $t_* \propto L_y$. For a primary field of dimension Δ , write $z' = \alpha_{\text{st}} t + iy$. The cylinder correlators have spatial and temporal factors $\sin(\pi \Delta y / L_y)$ and $\sinh(\pi \alpha_{\text{st}} t / L_y)$:

$$g_{\text{sp}}(\Delta y, 0) = \left(\frac{\pi}{L_y} \right)^{2\Delta} \left[\sin \left(\frac{\pi \Delta y}{L_y} \right) \right]^{-2\Delta}, \quad (27)$$

$$g_{\text{tm}}(0, t) = \left(\frac{\pi}{L_y} \right)^{2\Delta} \left[\sinh \left(\frac{\pi \alpha_{\text{st}} t}{L_y} \right) \right]^{-2\Delta}. \quad (28)$$

At $\Delta y = L_y/2$, the sine is unity. Matching the spatial and temporal factors cancels Δ and gives [3]

$$\alpha_{\text{st}} = \frac{\text{arsinh}(1)L_y}{\pi t_*} = \frac{\log(1 + \sqrt{2})L_y}{\pi t_*}. \quad (29)$$

The subscript distinguishes this spacetime conversion from the circuit parameters α_1 and α_2 .

The numerical proxy is mutual information between two spectator references. For each of the two canonical modes of a wall cell, first measure its occupation m , append a reference mode in occupation $1 - m$, and apply the number-conserving beam splitter

$$\begin{pmatrix} \hat{f}' \\ \hat{r}' \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \hat{f} \\ \hat{r} \end{pmatrix}. \quad (30)$$

Each pair contains one fermion and has reference entropy $\log 2$ for either outcome. One reference subsystem therefore contains two modes and has fresh entropy $2 \log 2$; two

TABLE I. Worked coarse-audit point estimates. These arithmetic means and interpolations are an estimator example, not calibrated values of t_* .

equil.	I_{sp}	temporal bracket	interpolated t_*
$2L_y$	0.32858	$I(1) = 0.51091, I(3) = 0.20135$	2.178
$3L_y$	0.12656	$I(6) = 0.19559, I(9) = 0.07446$	7.710
$4L_y$	0.28096	$I(1) = 0.44209, I(3) = 0.20006$	2.331
$6L_y$	0.22691	$I(1) = 0.47091, I(3) = 0.18730$	2.721

insertions append four spectator rows to the live occupied frame. Subsequent controller measurements have support only on the original physical rows, but physical-reference correlations continue to evolve through the common frame. If ζ_a are eigenvalues of a reduced reference correlation matrix,

$$S(A) = - \sum_a [\zeta_a \log \zeta_a + (1 - \zeta_a) \log(1 - \zeta_a)],$$

$$I(R_1:R_2) = S(R_1) + S(R_2) - S(R_1 R_2). \quad (31)$$

Only 2×2 and 4×4 matrices are diagonalized. The late-time matching point is defined operationally, before any curve fit, by

$$I_{\text{tm}}(t_*) = I_{\text{sp}}(L_y/2). \quad (32)$$

It asks how many circuit cycles give the same effective separation as halfway around the edge. Equation (29) converts this point under the CFT cylinder hypothesis; the matching equality itself is not evidence for CFT behavior.

The $L_y = 24$ CPU audit evaluates the matching estimator directly, not by fitting Eqs. (27)–(28). For each $m = 2, 3, 4, 6$, 32 complete Born trajectories are equilibrated to mL_y . From each reused checkpoint, R_1 is inserted at a sampled wall cell. The spatial branch inserts R_2 simultaneously at separation $L_y/2 = 12$; the temporal branches insert it at the same cell after $\Delta t \in \{1, 3, 6, 9, 12, 15, 18\}$. After the second insertion the same branch is followed for $3L_y = 72$ cycles. Relative cycles 49–72, the final $L_y = 24$ values, are averaged first within each trajectory and then arithmetically over the 32 trajectories. The estimator requires the shortest temporal point to exceed the spatial mean and takes the first adjacent downward crossing on the ordered sampled grid, linearly interpolated; global monotonicity is not assumed.

For example, at equilibration $2L_y$, the saved coarse means are $I_{\text{sp}} = 0.32858$, $I_{\text{tm}}(1) = 0.51091$, and $I_{\text{tm}}(3) = 0.20135$. Hence

$$t_* = 1 + \frac{0.51091 - 0.32858}{0.51091 - 0.20135}(3 - 1)$$

$$= 2.178. \quad (33)$$

Applying the identical operation to all completed coarse checkpoint files gives Table I.

The $3L_y$ row illustrates sensitivity to the measured spatial baseline: its unusually low I_{sp} moves the first crossing from the 1–3 interval to 6–9. Table I excludes

refinement points $\Delta t = 2, 7, 8$ and the paired trajectory bootstrap and convergence gates. It therefore carries no uncertainty interval and is not converted into a reported α_{st} . Evidence for the CFT interpretation would additionally require collapse of full spatial and temporal curves under this rescaling at multiple L_y .

VI. DISTINCTIONS

The three calculations answer different questions. G4 uses a maximally mixed input to reconstruct the absolute singular spectrum of a realized mEO, including its normalization and charge-resolved many-body tower. G5 uses independent pure trajectories to estimate the Grassmannian Jacobian of the normalized covariance dynamics over a late finite-time window. The reference calculation instead compares a state observable at two spacetime separations to calibrate the conversion between circuit cycles and the spatial cylinder. The frame/Möbius equivalence relates two coordinate descriptions of G5; neither it nor the max-mix endpoint identity collapses these three estimators into one object.

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